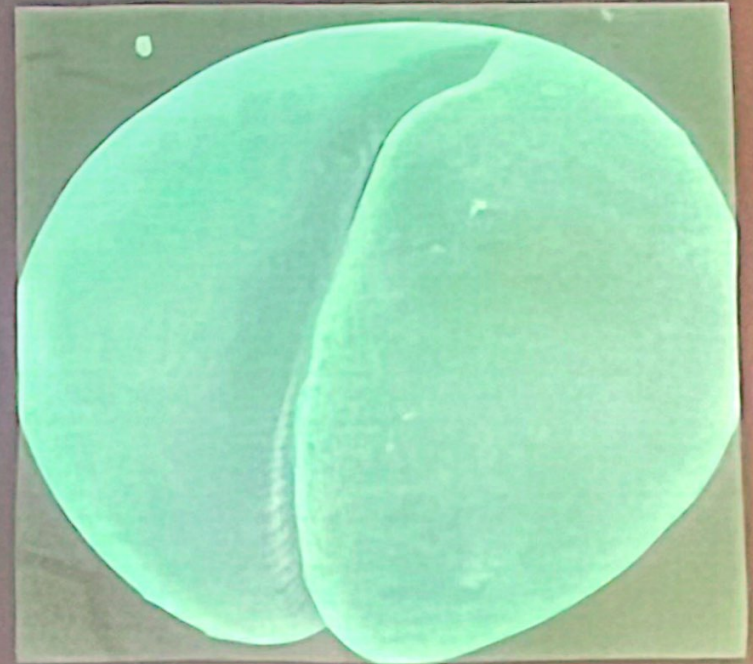


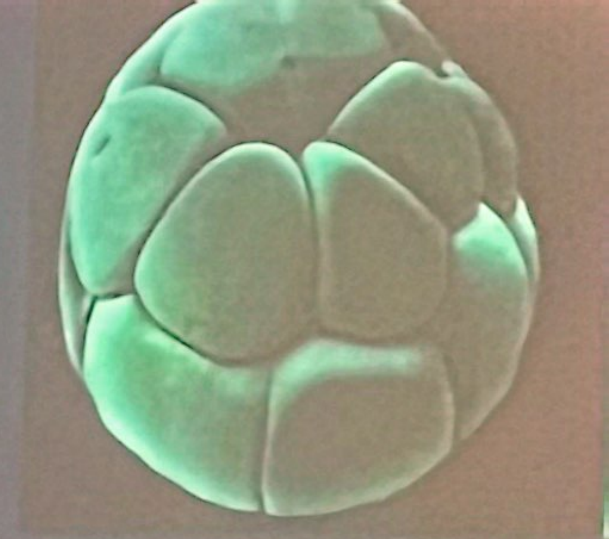
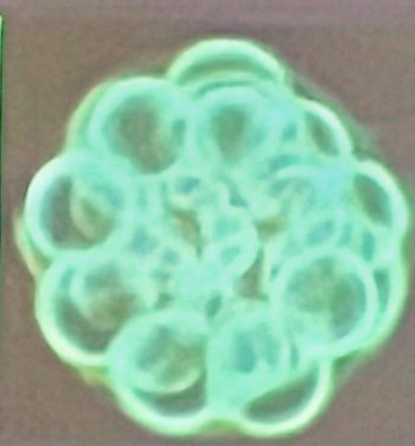
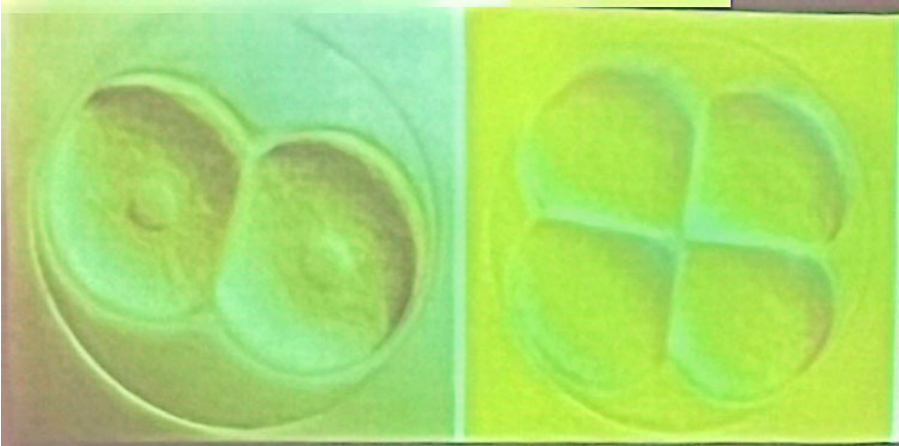
Cleavage (Segmentation)

Successive Mitotic division of the zygote (convert one cell to multicellular, morula and blastula inside the zona pellucida (at uterine tube)

It is characterized by:

- 1-Increase number of cells
(blastomeres) decrease at size
- 2- Double sequences (2,4,8,16,....)
- 3- Formation the morula
- 4- Appearance of cavity (blastocoel)
So called blastula





Morula:

Formation of compact ball of cells

Varied in cell numbers at mammals

8-cells in human

16- cells in sheep

32-cells in cattle-

❑ This ideal stage of embryo transfer

Types of cleavage

Depend on the type of ova ,amount and distribution of yolk

In general yolk retards completion of cell separation(cytokinesis)

1- Total(Holoplastic) cleavage: zygote divides completely into blastomeres.(cytoplasm+nucleus+yok)

Equal: blastomeres equal in amphioxus & mammals.

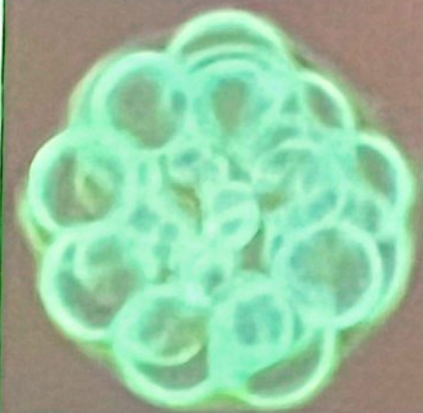
Unequal: blastomeres unequal in frog and fish.

2- Partial (meroblastic)cleavage: zygote divides in animal pole only.(cytoplasm +nucleus)

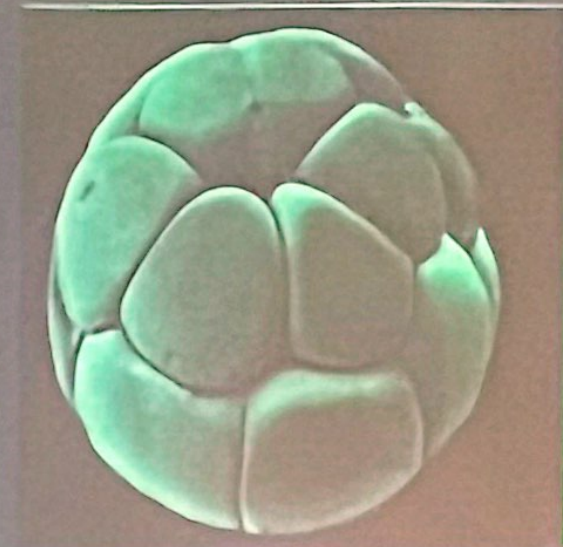
Discoidal: the division in germinal disc in birds and reptile

Superficial: the division in peripheral cytoplasm in flies.

Total equal
cleavage



Holoblastic unequal where there
is more yolk at the vegetal pole.
Cleavage occurs more rapidly in the
animal pole than in the
vegetal pole.



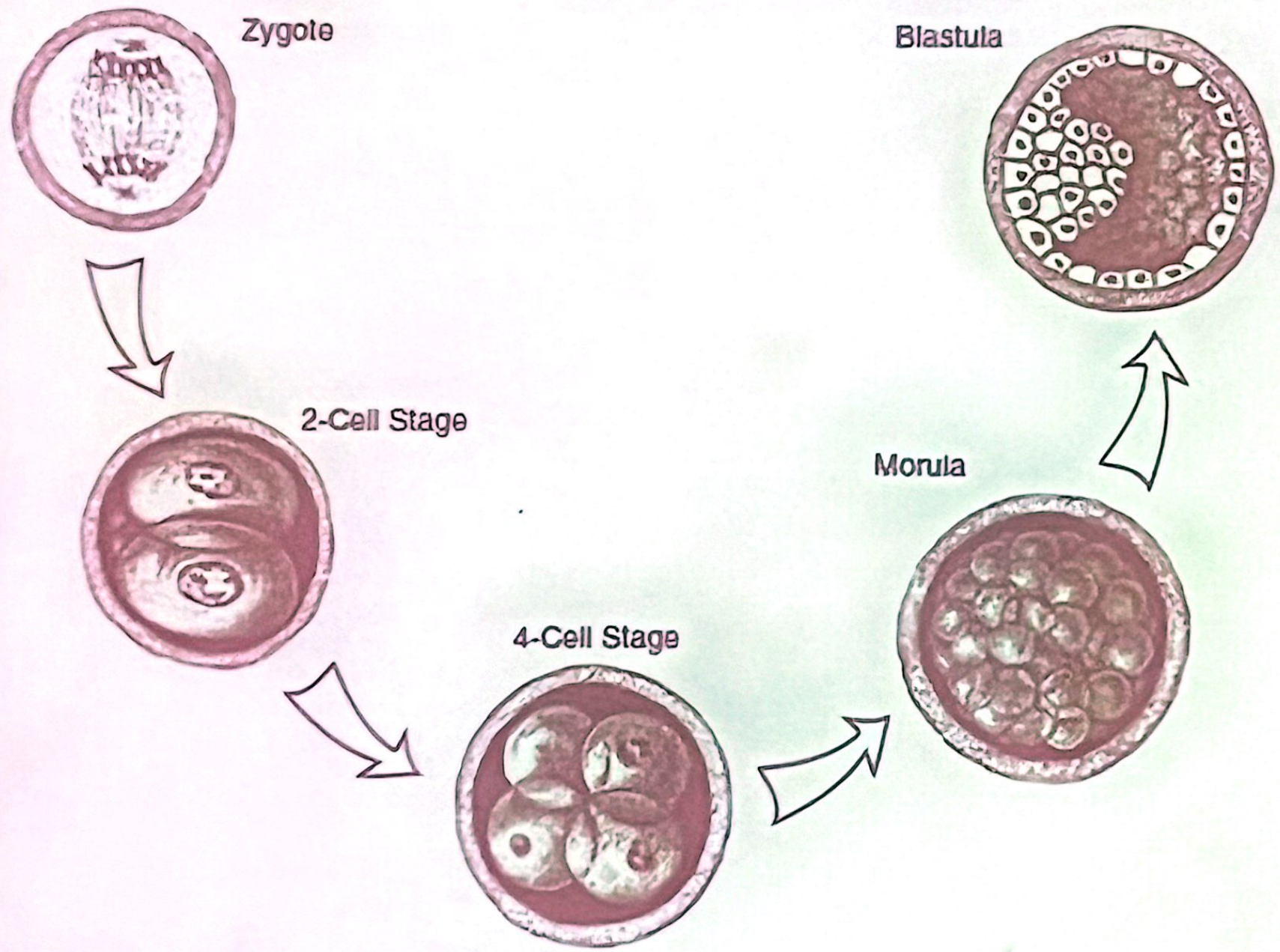
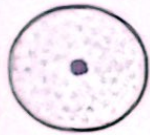
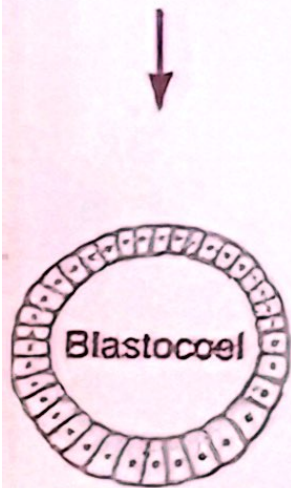


Figure 3-1. Development from zygote to blastula.

Isolecithal egg

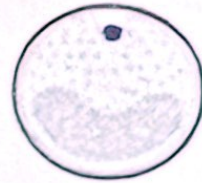


↓
Holoblastic
cleavage

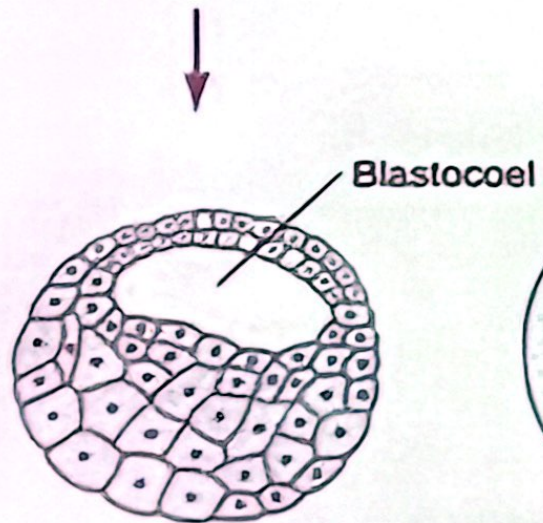
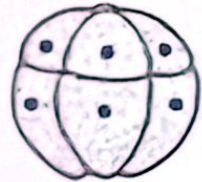


Sea urchin

Mesolecithal egg

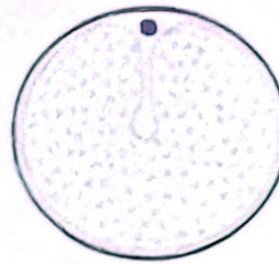


↓
Holoblastic
cleavage

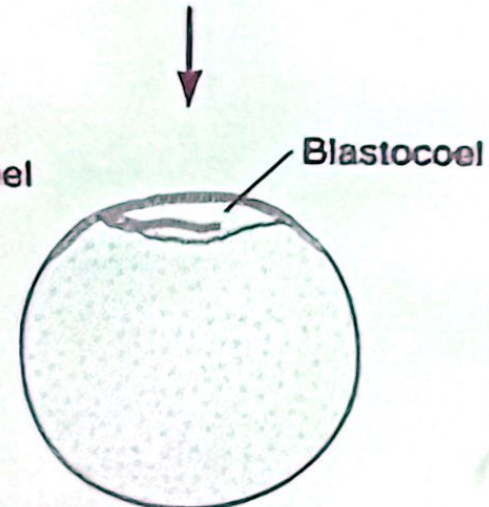
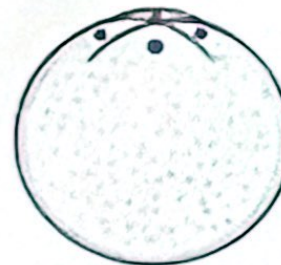


Frog

Telolecithal egg



↓
Discoidal
cleavage

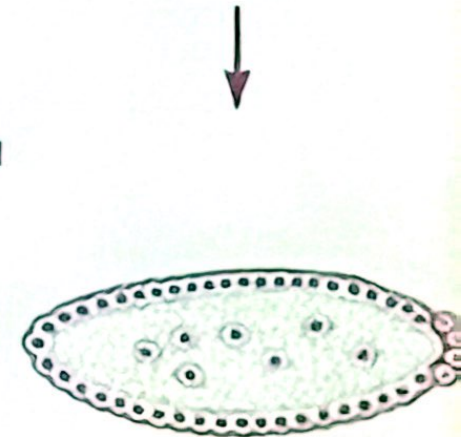
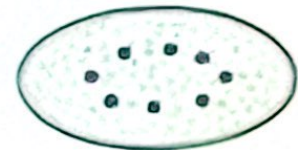


Bird

Centrolecithal egg

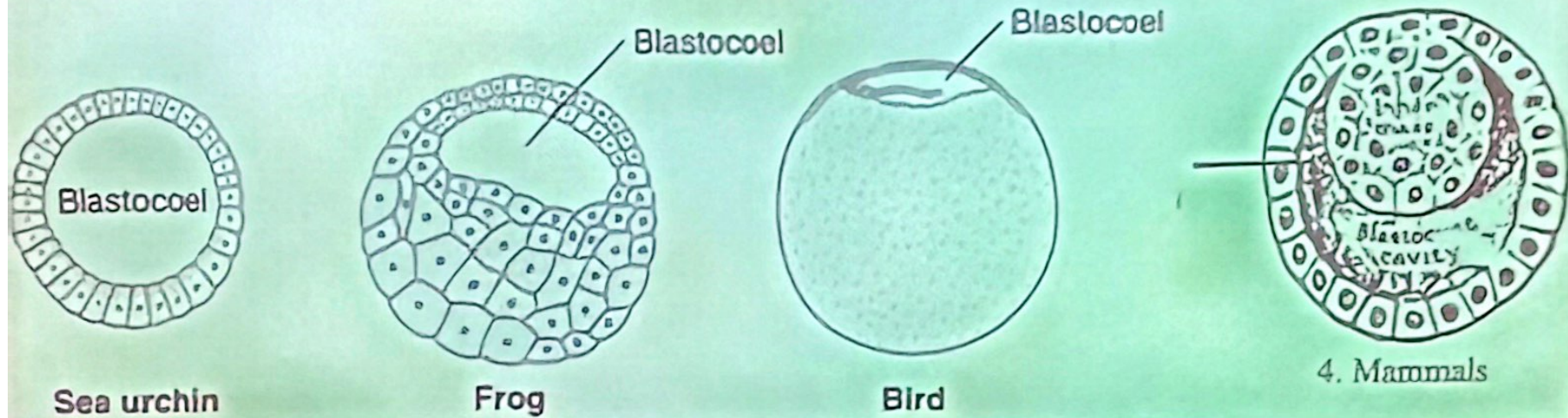


↓
Superficial
cleavage



Insect

Types of blastula



in amphioxus: The blastula is a hollow sphere consists of one layer of equal cells. The blastocoele is in the center and filled with fluid.

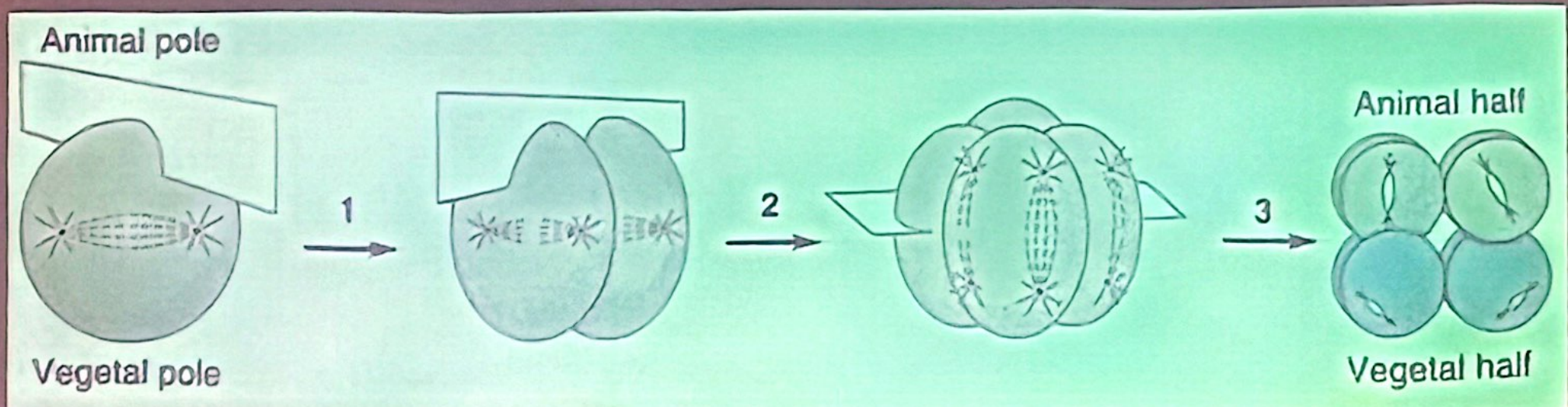
in amphibian: The blastula wall consists of more than one layers of unequal cells (micromere and macromere). The blastocoele is eccentric.

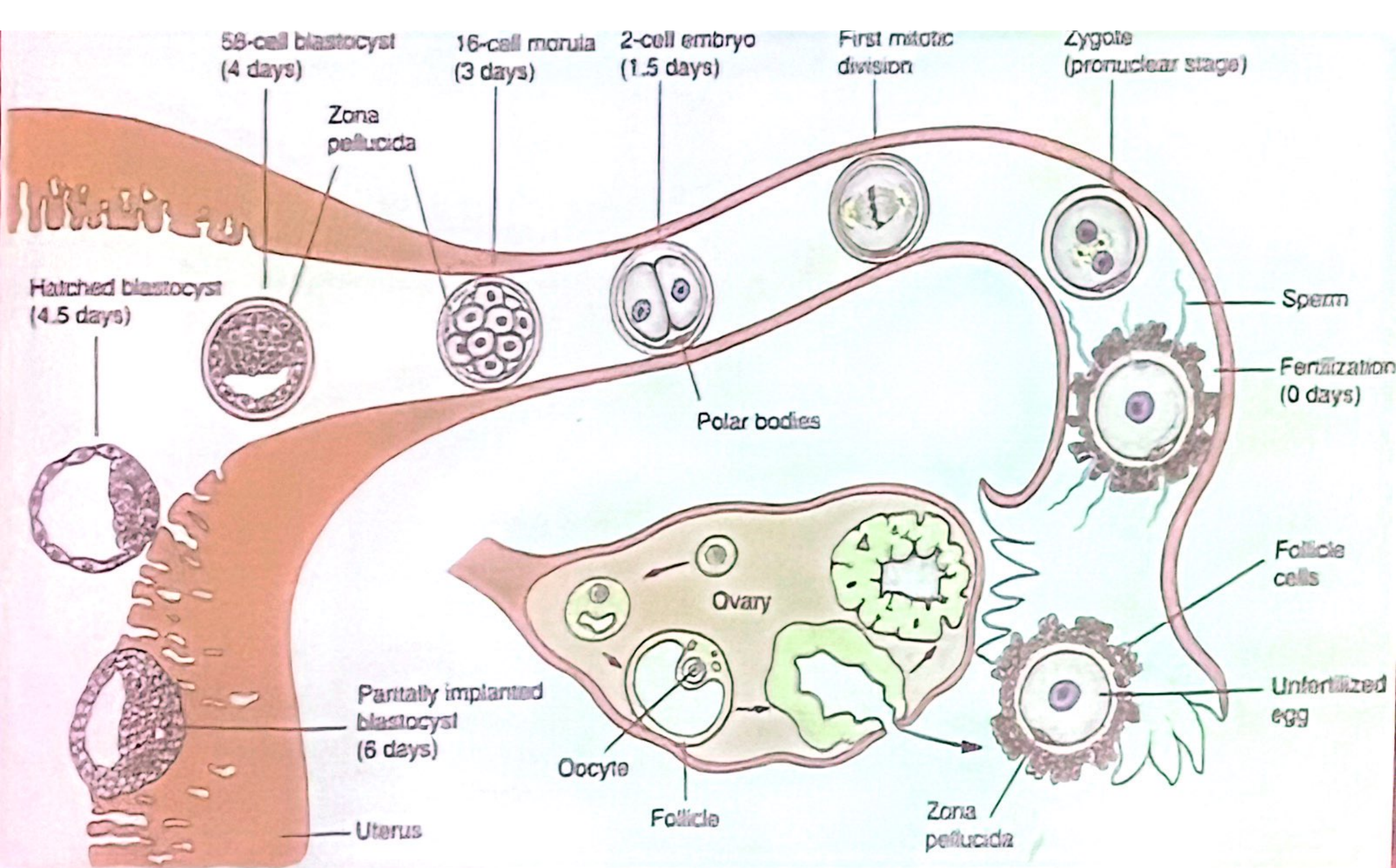
in birds: The blastula is disc divided into two zones, zona opaca in the prephery and zona pellucida in the center.

in mammals: The blastula consists of outer layer is trophoblast and inner cell mass.

Cleavage in amphioxus

- The cleavage is total and equal.
- The plane 1st **vertical**- 2nd **meridional** (from pole to pole at the right angle with the first - 3rd is **horizontal** and 4th division is **meridional** till form the **morula**, 5th, 6th, 7th, **vertical** till form
- The **blastula** is a hollow sphere consists of one layer of equal cells, blastocoel in the center and filled with fluid





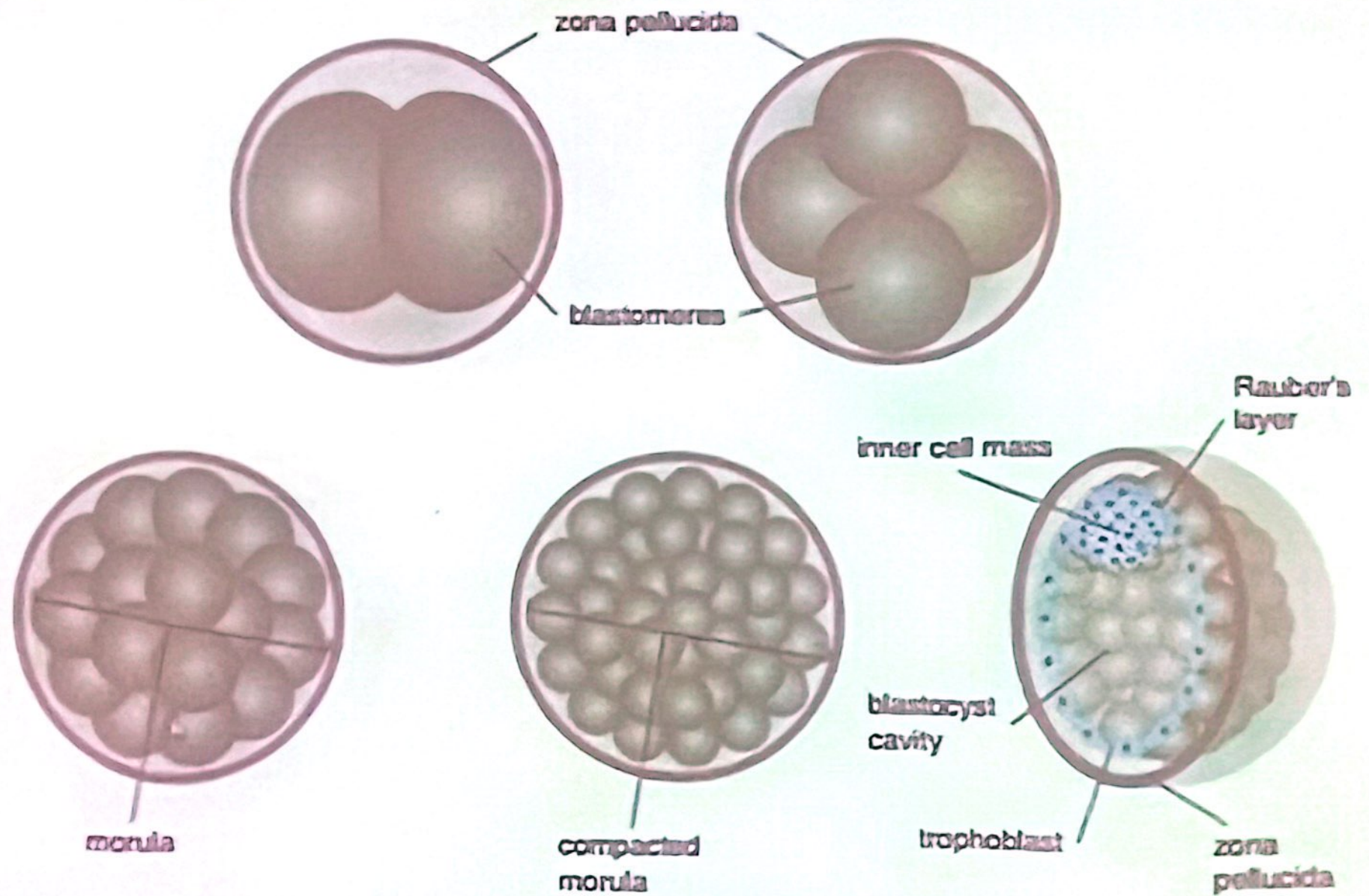
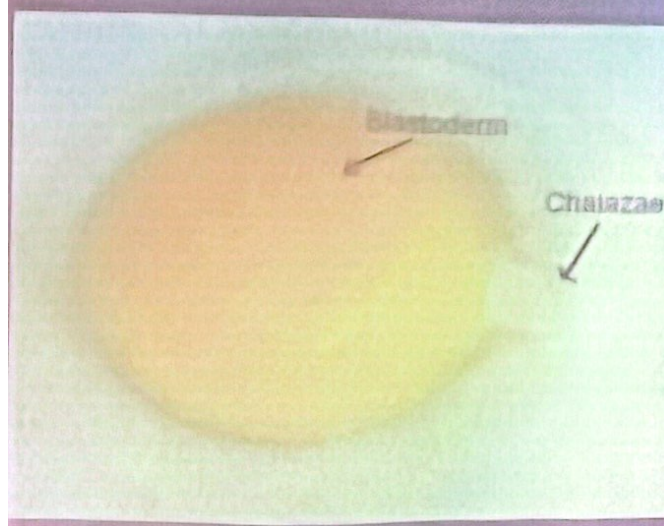


Figure 5.3 Stages of cleavage in a mammalian embryo from the two-cell stage to the formation of a blastocyst.

Cleavage occurs at blastoderm
small whitish circular area



cleavage occurs only in the blastodisc,

A small disc of cytoplasm 2–3 mm in diameter

1st vertical 2nd meridional , 3rd radial ,4th

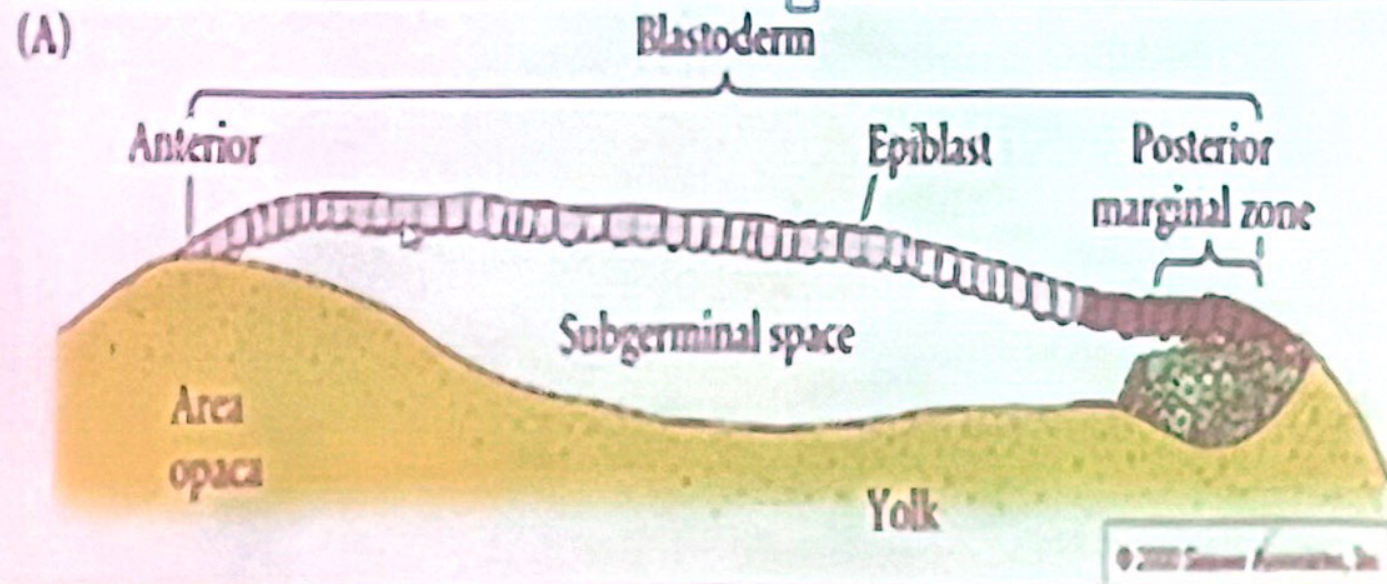
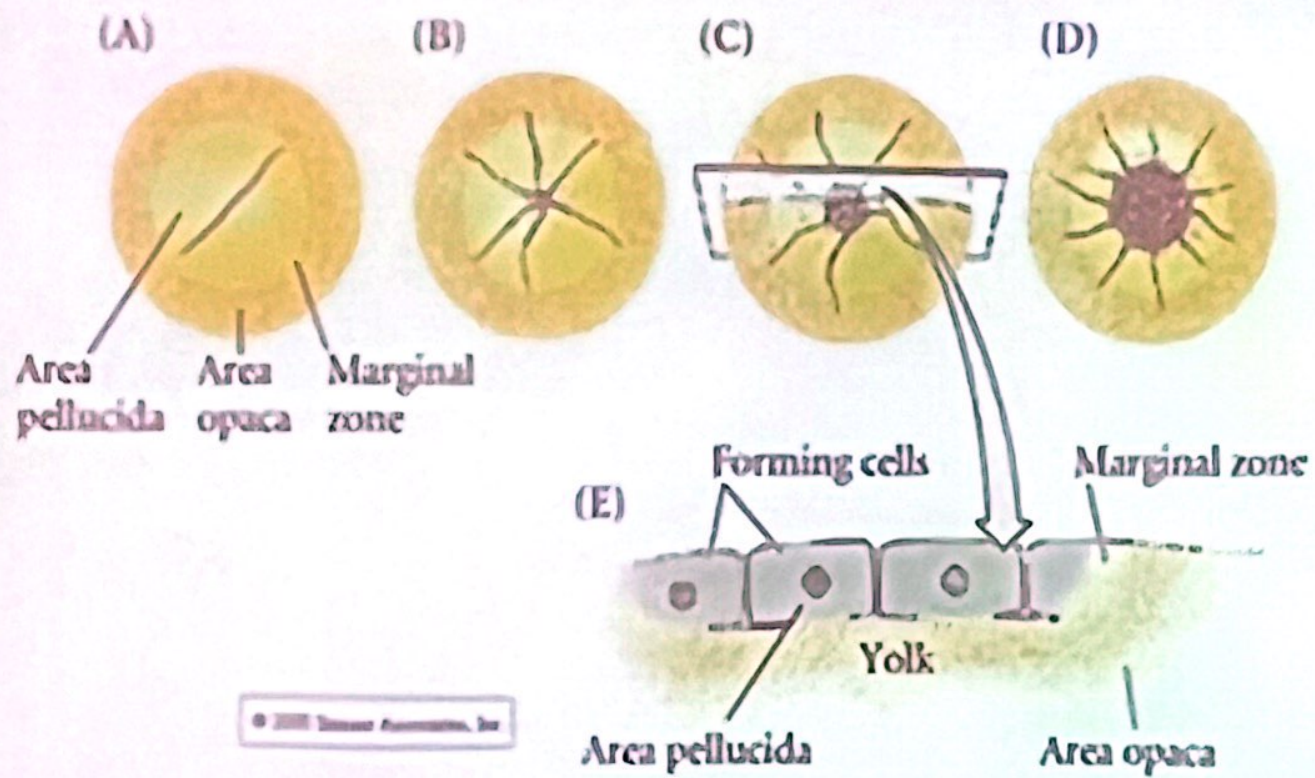
circumference (single layer of cells)

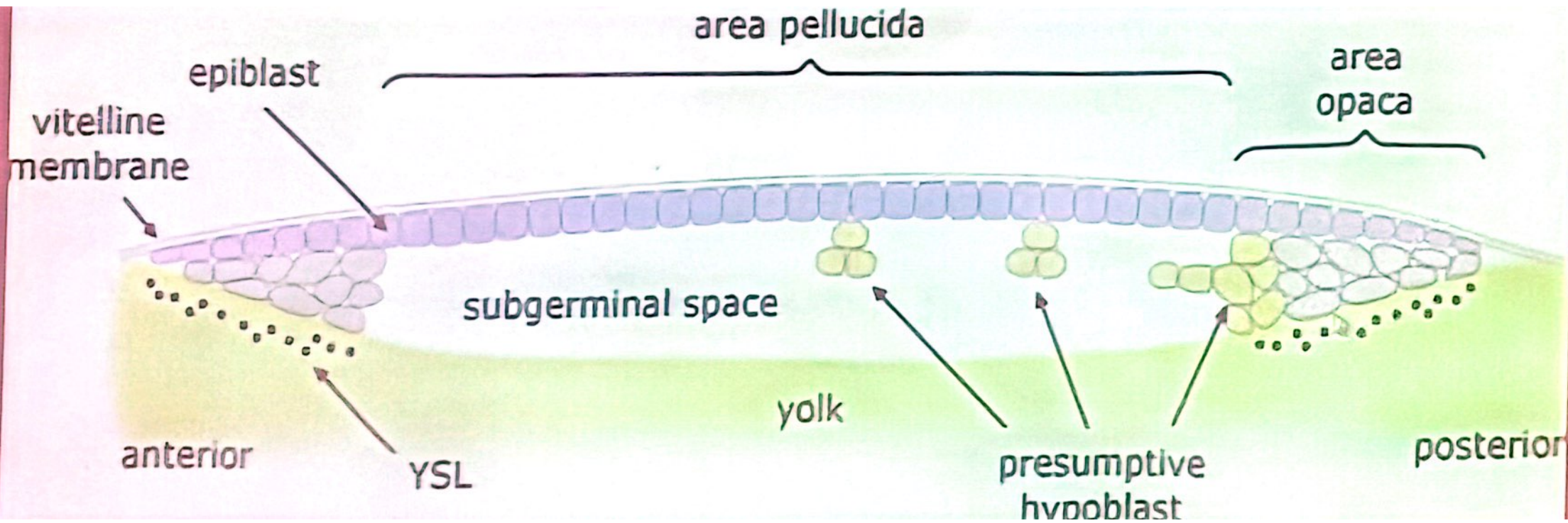
Later cleavages divide the blastoderm into a
5–6 cell layers thick tissue.

Between the blastoderm and the yolk is a
space called the **subgerminal space**

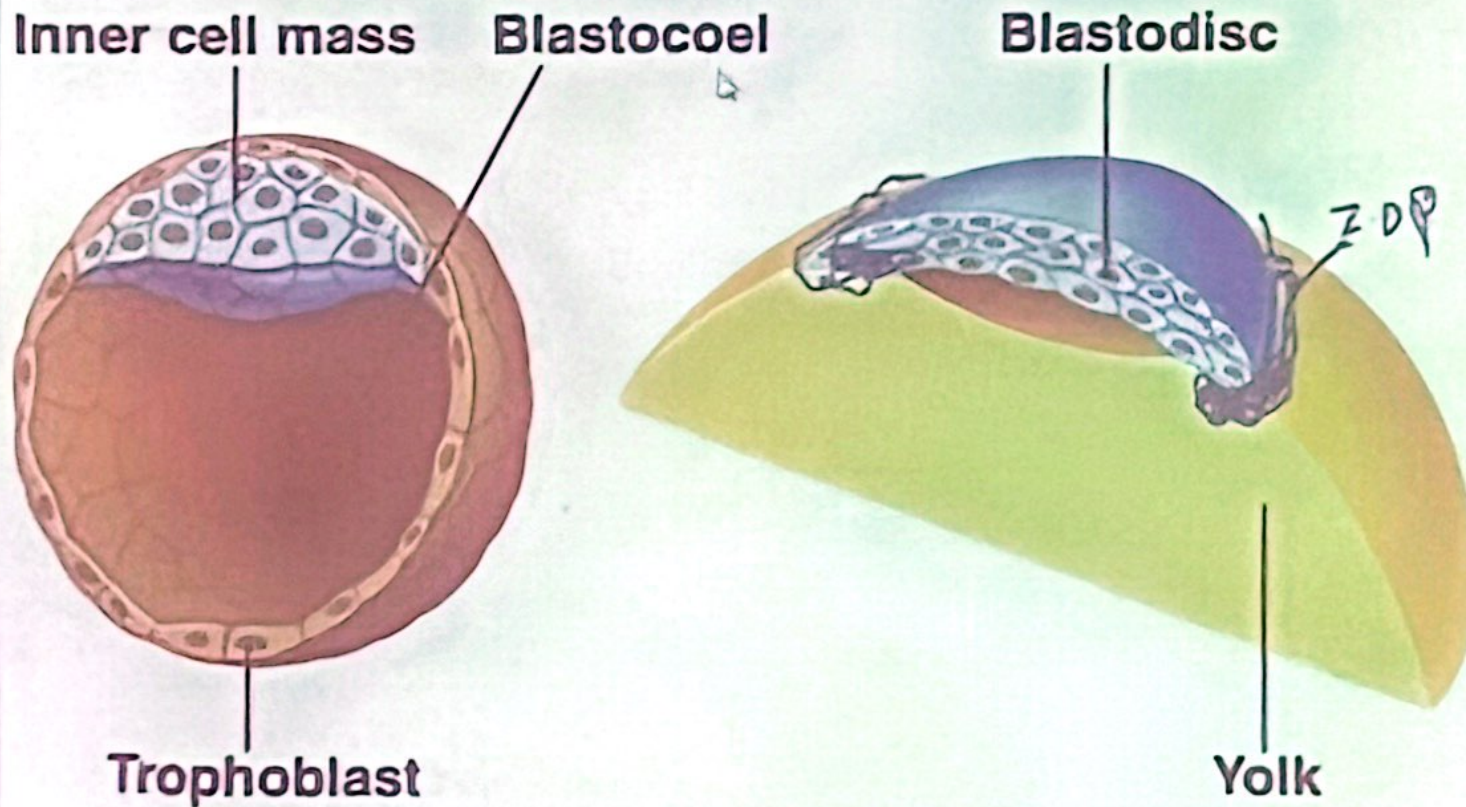
area pellucida : part of the blastoderm that
forms most of the embryo proper

The peripheral blastoderm cells retain their
deep cells and constitute the **area opaca**.





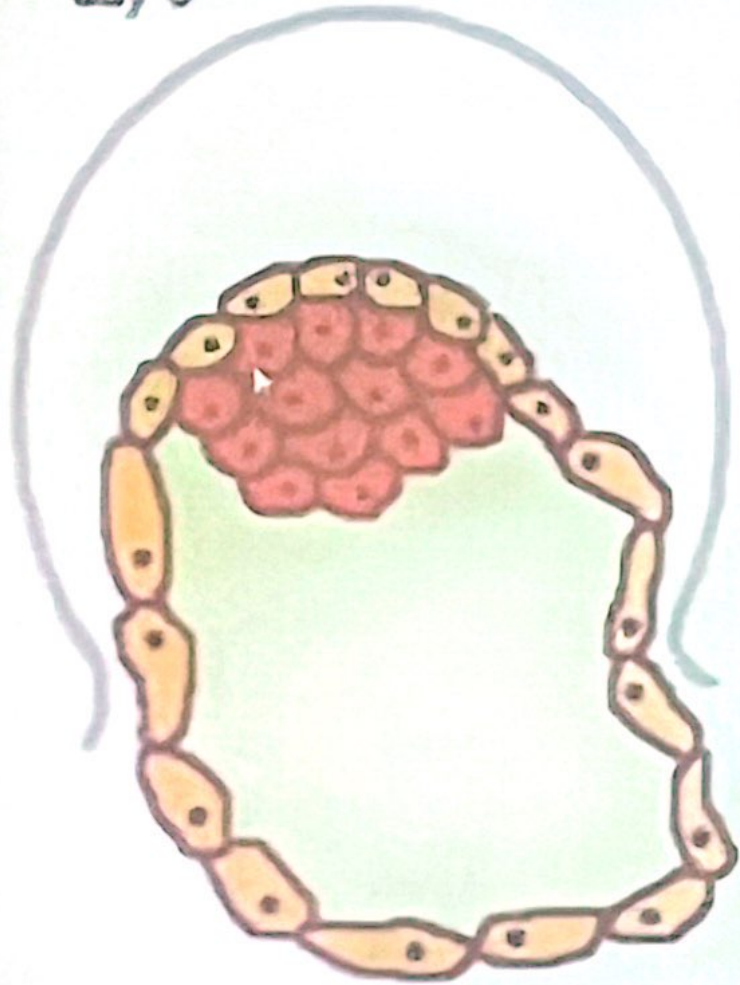
Blastula of Mammals and Birds



- Cap of cells develops into a blastodisc
- Blastocoel develops in mammals, surrounded by trophoblast

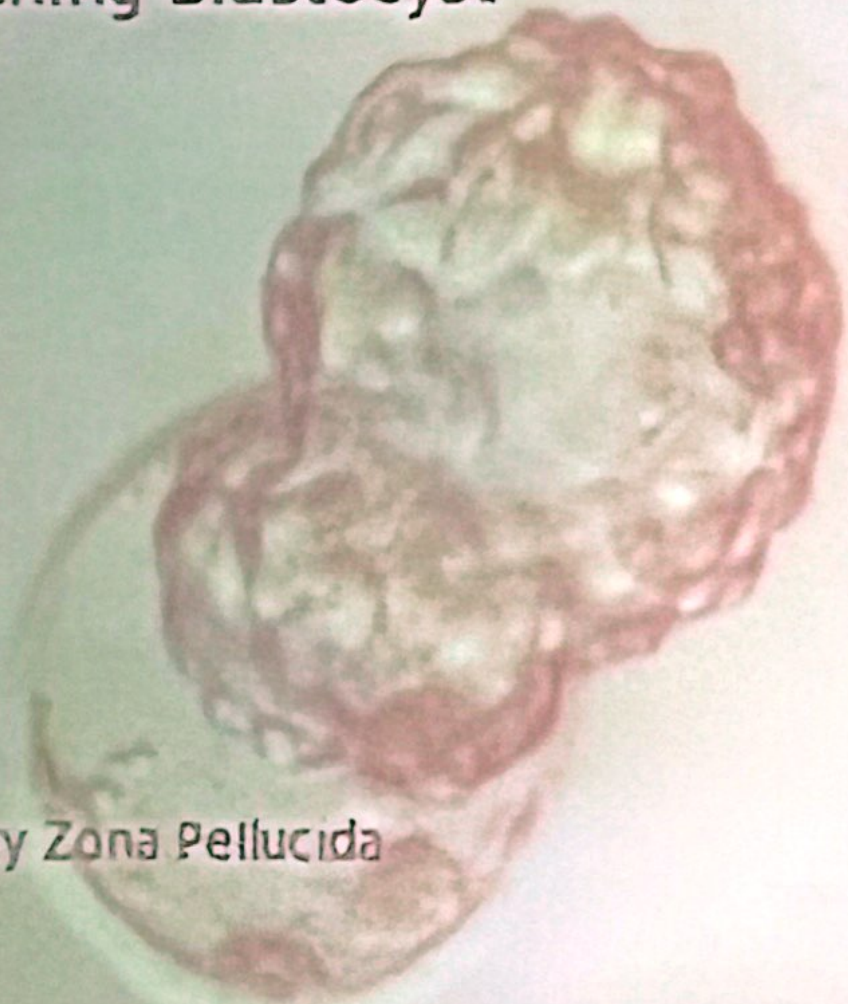
6

day 6



zona hatching

Hatching Blastocyst



Empty Zona Pellucida

Photo courtesy of Robert Shabanowitz, PhD
Geisinger Medical Center.

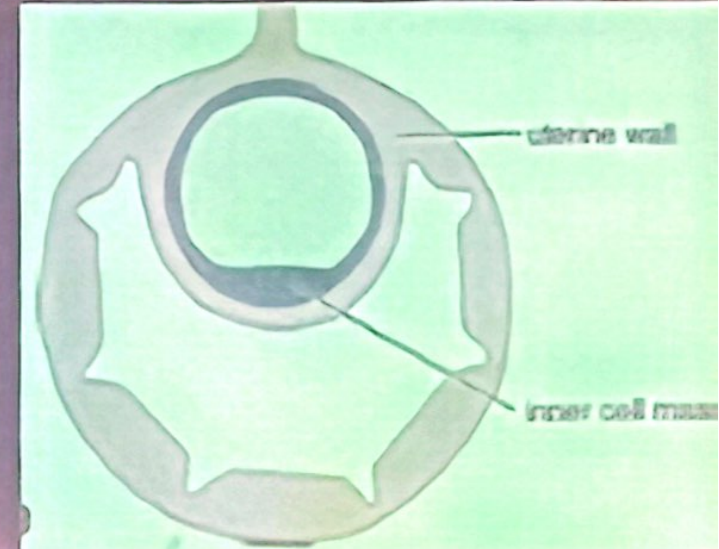
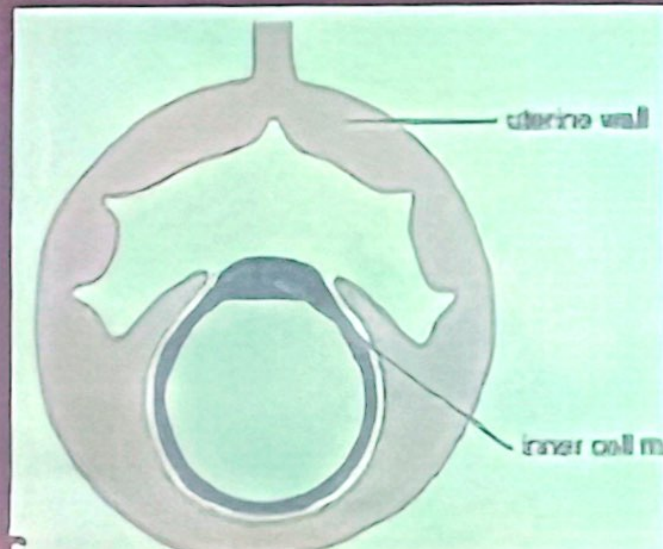
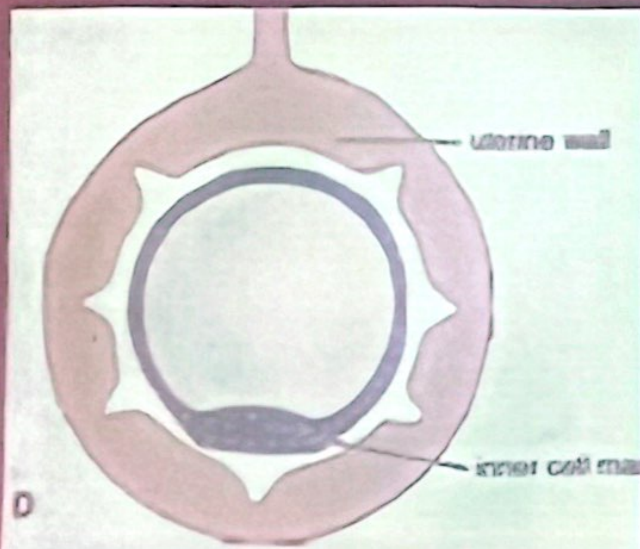
Implantation

It is interaction process between blastocyst and uterine mucosa(endometrium)

superficial (equine –ruminant-pig)

Eccentric; carnivores- rodent

Invasive: human



Membrane of ovum

primary

loc.

tertiary

vitelline
membrane

Corona radiata

↑
Zona pellucida

Jelly
=
rag

alb.
robbile

alb.
↑
shell
husk

Gastrulation

blastula move form the three germ layers:
ectoderm, mesoderm, and endoderm
(In a gastrula [Gr.= little stomach] one can see evidence
of primitive gut formation.)

General characters:

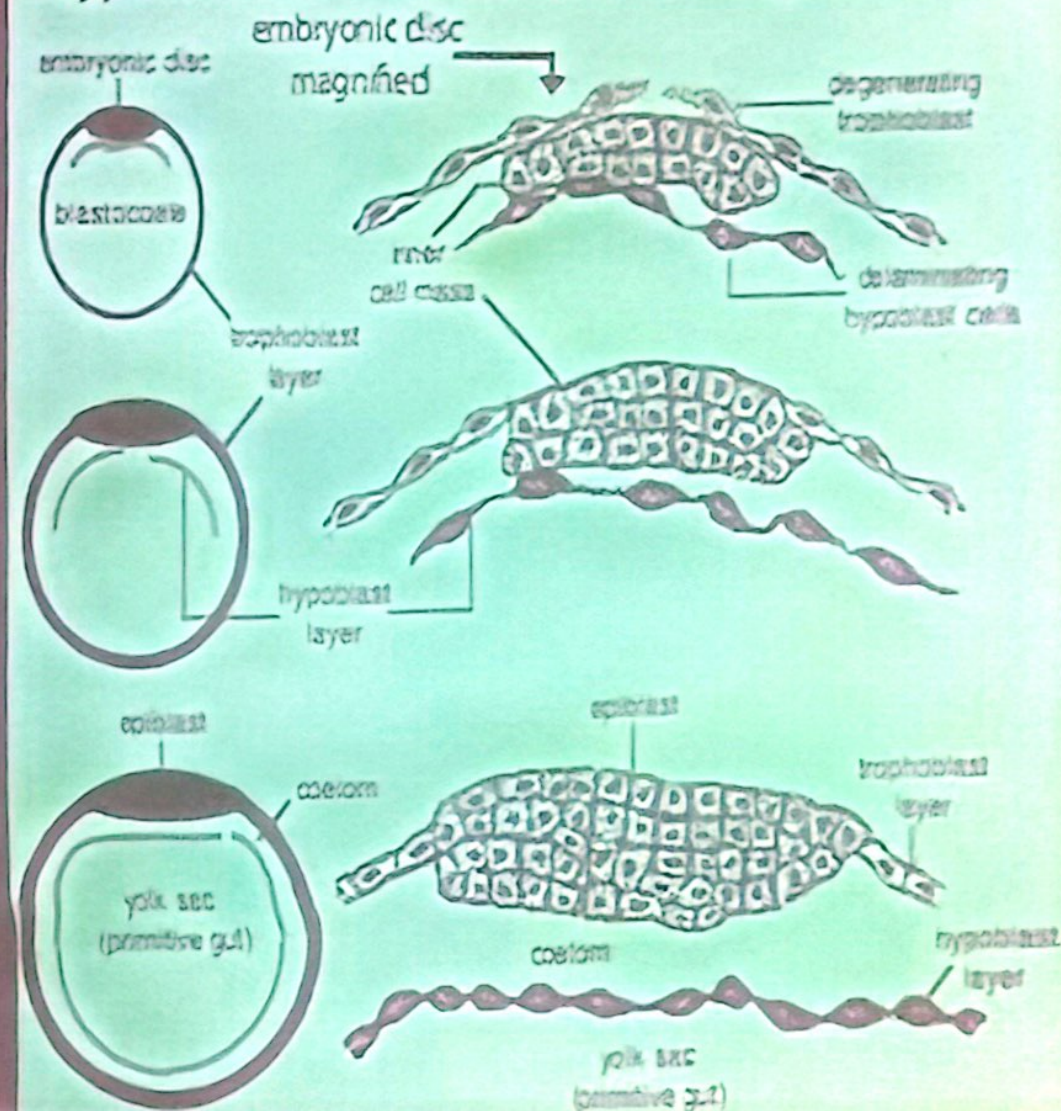
- 1-Cell division becomes slower & Cell migration increase
- 2- Appear formation of primitive streak
- 3- appearance of notochord
- 4- 3 germ layers formation
- 5- Disappearance of P.S. is finishing of gastrulation



From the inner cell mass, cells proliferate, break loose (**delaminate**), and migrate to form a new cell layer **hypoblast**; it forms a **primitive yolk sac**. The remaining inner cell mass may henceforth be called **epiblast**.

On the epiblast surface, a **primitive streak** forms as differential cell growth generates a pair of ridges connected by a depression.

Hypoblast Formation (three stages)



Trophoblast

Amnion

Embryonic disc:

Epiblast Ectoderm

Hypoblast Endoderm

Yolk sac

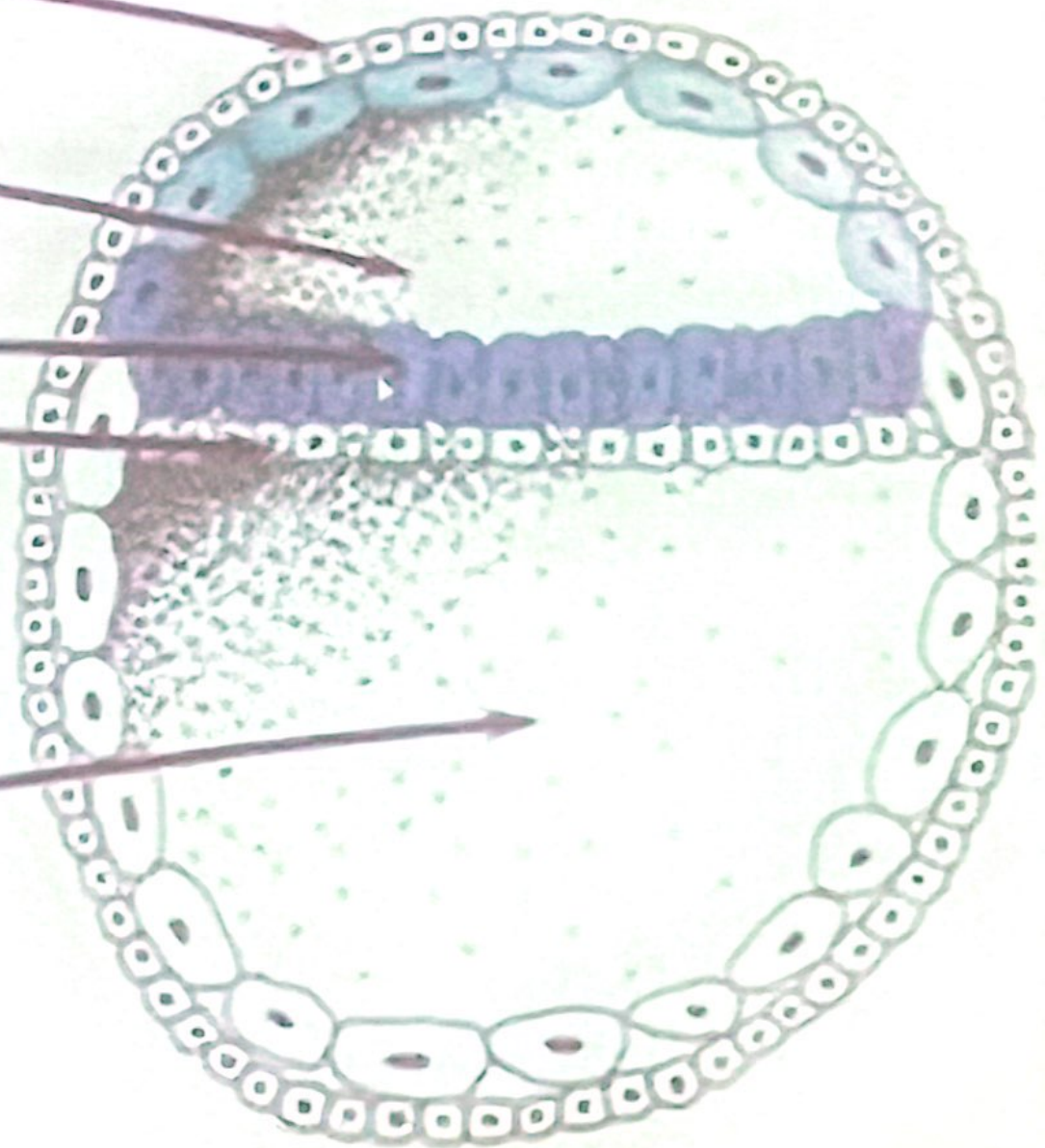




Fig. 7-2: Accumulation of epiblast cells in the posterior portion of the embryonic disc results in the formation of the primitive streak (1).

Hensen's node

primitive streak

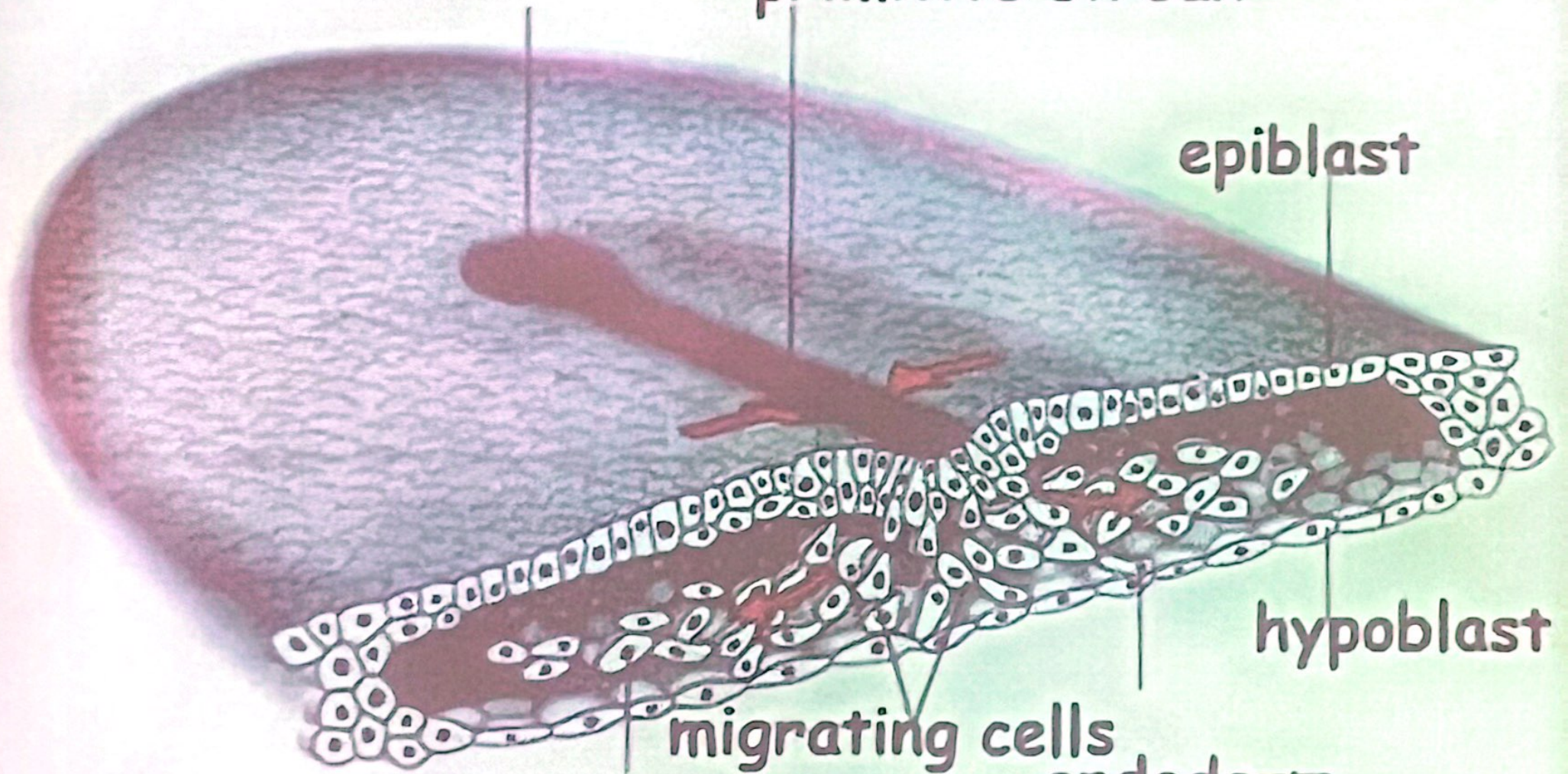
epiblast

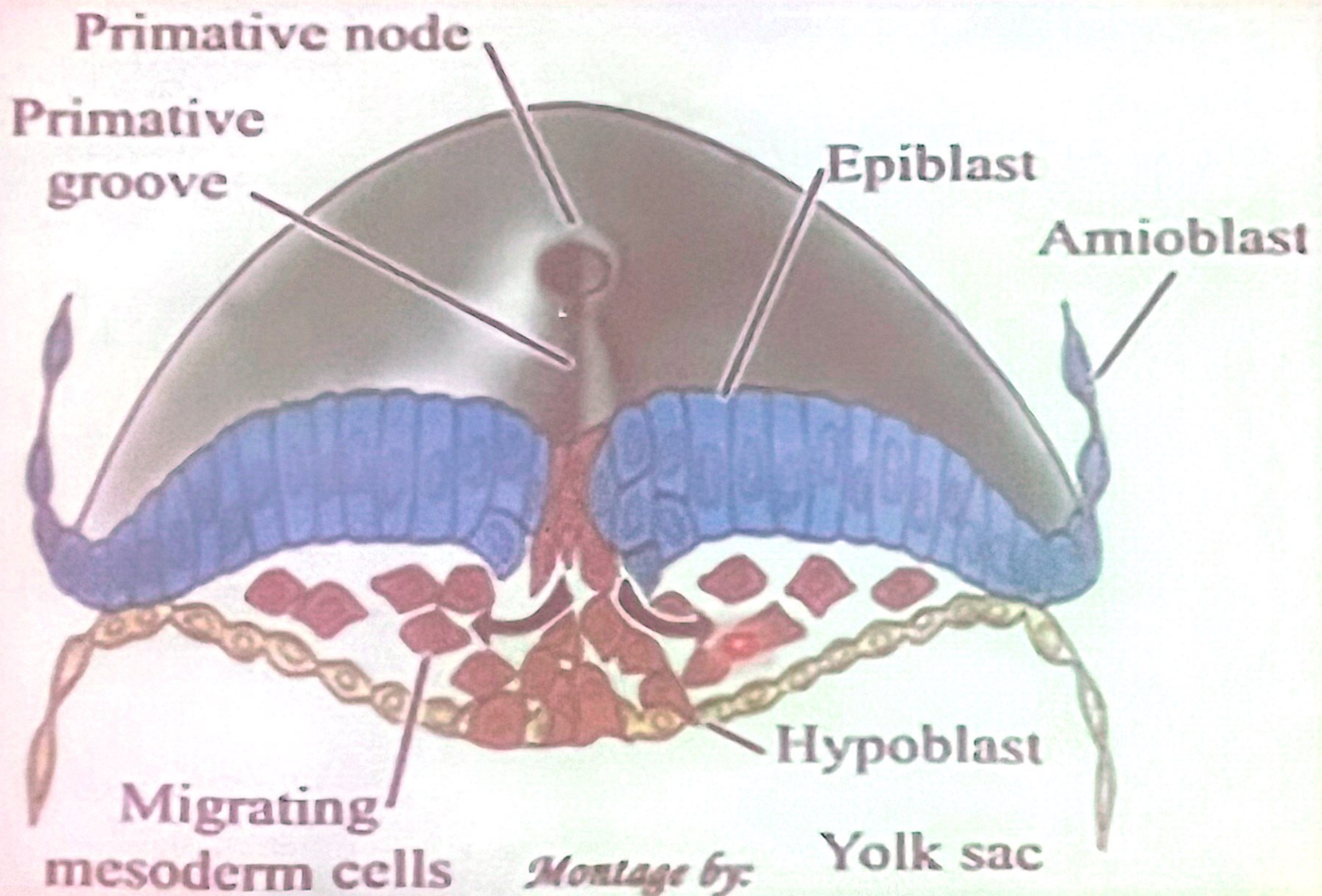
hypoblast

migrating cells

endoderm
displacing
hypoblast

mesoderm





- The **notochord** develops from the primitive node located at the cranial end of the primitive streak. From the node, cells proliferate and migrate forward into the future head region where they become the rod-shaped
- Act as primitive Skelton

